

NATHAN MANLEY

Cork, Ireland | First-Year Engineering Student at UCC | Embedded Systems and Space Enthusiast
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More information on my projects and experience available on my website.

EDUCATION

University College Cork — 1st Year Engineering — First Class Honours (1.1) | GPA: 3.75/4 2025 - 2029

Quercus Talented Student Programme Innovation & Entrepreneurship Scholar — Fully funded scholarship, awarded as recognition for entrepreneurial and innovative endeavours both on and off campus.
2026 - 2029

Glanmire Community College — Leaving Certificate — 589/625 Points 2025

Subjects: Applied Maths, Physics, Technology, Maths, English, History, Irish, French

WORK EXPERIENCE

Logitech — Software UX Design Intern Summer 2026

- Designing UX elements for future Logitech Product with Software POVs.

ATMac — Repair Technician Summer 2023/2024

- Diagnosed and repaired various Apple laptops and desktops.
- Enhanced technical troubleshooting and component level repair skills.

TECHNICAL / ENGINEERING EXPERIENCE

National SDG Hackathon

- Winner of National SDG Hackathon, UCC and Best Pitch Award, winning €4K in prize money.
- Developed Remento, an app for early-stage dementia tracking/detection for families of the elderly.

HackEurope 2026

- Developed WindSite, utilising Claude, Python, and React Web Development.
- WindSite is an intelligent platform that integrates a modular node system flow chart to make offshore wind turbine farm planning easier and smarter.

Patch 2024

- Selected as one of 30 innovators across Europe to complete the program in Dublin.
- Co-founded Gymificient, a platform providing actionable analytics to gym owners for machine acquisition and maintenance.
- Developed backend logic and analytics modules using OpenCV, in C++ and Python.

UCC Rocketry and Space Exploration Society — Payload and Hardware Avionics Team Member 2025–2026

- Programming camera telemetry and payload systems for MACH '26. AV Bay 3D Designer for EuRoC '26.

UCC Claude Code Club — Workshop Tutor 2026–2027

- Designing interactive workshops for those new to AI development and use, teaching them how to use Claude tools intelligently and ethically.

CanSat — Lead 3D Can Design 2024

- Designed core satellite structure using Autodesk Fusion 360.
- Optimized housing for electronics and produced parts on my personal 3D printer.

CanSat — Lead Programmer 2023

- Implemented real-time telemetry and data processing in C++.
- Built post-processing algorithms and visualized results for presentation in the competition.

INDEPENDENT PROJECTS / ACHIEVEMENTS

- Telemetry Parsers: Captures and interprets UDP data from simulators like Gran Turismo 7 for integration with microcontrollers and dashboards. Various modules were programmed in C++, Python and Javascript.
- School Awards: Environmental and Innovation "Diograis" awards while attending GCC.

VOLUNTEERING

CoderDojo Glanmire

Joined as a programmer in 2014, receiving free mentorship. I became a mentor in 2023 to volunteer my time to teach today's young learners C++, Python, and HTML.

Gaisce

2023

Bronze Gaisce award.

INTERESTS

- Technical: Microcontrollers (ESP32/ESP8266), Embedded Systems, 3D Printing (Ender 3 V2).
- Personal: Weightlifting, Formula 1, Drifting.
- Academic: Electrical, Electronic & Mechanical Engineering. Rocketry and CanSat.

References available upon request